

Total No. of Questions : 6]

SEAT No. :

PC-1958

[Total No. of Pages : 2

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T.E. (Computer Engg.) (Honors)
DATA SCIENCE AND VISUALIZATION
(2019 Pattern) (Semester - I) (310501)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagram. must be drawn wherever necessary
- 3) Figures to the right-indicate full marks
- 4) Assume suitable data, if necessary

- Q1) a)** State and explain how Naïve bays Theorem is used to solve classification problems. **[6]**
- b) What is the difference between regression and classification? Explain with example. **[6]**
- c) Write a note on (Any 3) **[6]**
- i) Partitioning Clustering
 - ii) Density-Based Clustering
 - iii) Distribution Model-Based Clustering
 - iv) Hierarchical Clustering

OR

- Q2) a)** What is clustering? explain K-means clustering algorithm. **[6]**
- b) Explain how simple linear regression is used for house prize prediction. (Assume the suitable dataset). **[6]**
- c) Briefly explain evaluation of association rules. **[6]**

P.T.O.

- Q3) a)** Write a note on the following [9]
- i) Gini Index
 - ii) Information gain
 - iii) Entropy

- b) What is a neuron? Explain the architecture of artificial neurons. [8]

OR

- Q4) a)** What is a decision tree? What are the advantages and disadvantages of a decision tree? [9]

- b) What is a feedforward neural network? Explain with suitable example. [8]

- Q5) a)** What are the challenges related to data visualization? [6]

- b) Explain dashboard design principles [6]

- c) What are the advance data visualization techniques? Explain any 2. [6]

OR

- a) Write a note on 'Display media for dashboard'. [9]

- b) Explain where and how bar-graphs, Scatterplots and histograms can be used to visualize the data. [9]

- Q6) a)** Explain entity -relationship (ER) Data modelling. [6]

- b) list the disadvantages of multi-dimensional data model? [6]

- c) Discuss the challenges of clustering High-dimensional data. [5]

OR

- a) What are the key steps in data modelling process? Enlist and explain. [6]

- b) Explain multidimensional data model with one example. [6]

- c) What do you mean by Principal Component Analysis? Explain with example. [5]

